

Program 9: Organize kubernetes pods using labels and namespaces

File structure:

namespaces.yaml

pods.yaml

namespaces.yaml

apiVersion: v1

kind: Namespace

metadata:

name: dev

apiVersion: v1

kind: Namespace

metadata:

name: prod

pods.yaml

apiVersion: v1

kind: Pod

metadata:

name: frontend-dev

namespace: dev

labels:

app: frontend

env: dev

spec:

containers:

- name: frontend

image: nginx

ports:

- containerPort: 80

apiVersion: v1

kind: Pod

metadata:

name: backend-dev

namespace: dev

labels:

app: backend

env: dev

spec:

containers:

- name: backend

image: nginx

ports:

- containerPort: 80

```
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-prod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-prod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

Start minikube

```
#minikube start
```

Check the status to verify all the services are running

```
#minikube status
```

Create the pod from YAML files

```
#kubectl apply -f namespaces.yaml
```

```
#kubectl apply -f pods.yaml
```

List all the namespaces

```
#kubectl get namespaces
```

List pods of prod and dev namespaces

```
#kubectl get pods -n dev
```

```
#kubectl get pods -n prod
```

```
#kubectl get pods -n dev -l app=frontend
#kubectl get pods -n dev -l app=backend
```

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/lab9
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ cd ../lab9
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl apply -f pods.yaml
pod/frontend-dev created
pod/backend-dev created
pod/frontend-prod created
pod/backend-prod created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get namespaces
NAME                STATUS   AGE
default             Active  18d
dev                 Active  19s
kube-node-lease     Active  18d
kube-public         Active  18d
kube-system         Active  18d
kubernetes-dashboard Active  9h
prod                Active  19s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get pods -n dev
NAME                READY   STATUS    RESTARTS   AGE
backend-dev         1/1     Running   0           23s
frontend-dev        1/1     Running   0           23s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get pods -n prod
NAME                READY   STATUS    RESTARTS   AGE
backend-prod        1/1     Running   0           26s
frontend-prod       1/1     Running   0           26s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get pods -n prod -l app=frontend
NAME                READY   STATUS    RESTARTS   AGE
frontend-prod       1/1     Running   0           35s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get pods -n prod -l app=backend
NAME                READY   STATUS    RESTARTS   AGE
backend-prod        1/1     Running   0           48s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$ kubectl get pods -l app=backend
No resources found in default namespace.
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab9$
```

Delete all resources

```
#kubectl delete pods --all -n dev
#kubectl delete pods --all -n prod
#kubectl delete namespaces dev prod
```